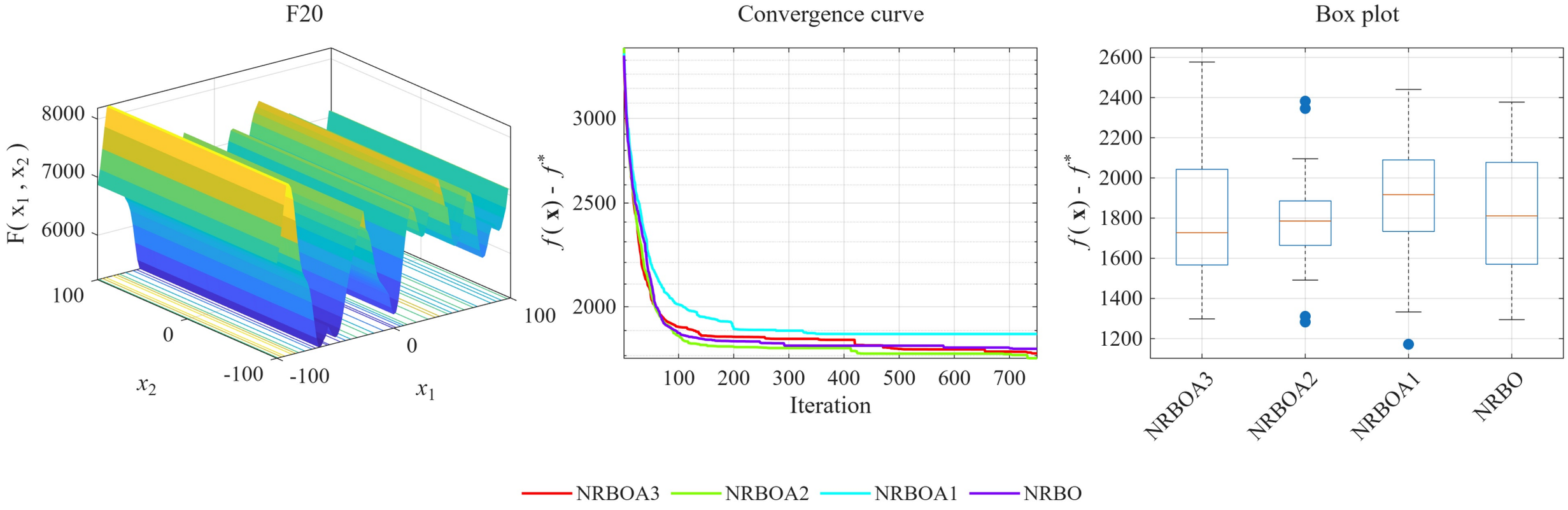

F20
F(x1 , x2)
8000
7000
6000
100
0
-100
x2
-100
0
x1
Convergence curve
f(x) - f*
3000
2500
2000
100
100
200
300
400
500
600
700
Iteration
Box plot
f(x) - f*
2600
2400
2200
2000
1800
1600
1400
1200
NRBOA3
NRBOA2
NRBOA1
NRBO
NRBOA3
NRBOA2
NRBOA1
NRBO